

```

bot face.py  datalogger.py
1 import tkinter as tk
2 from tkinter import ttk
3 import serial
4 import matplotlib.pyplot as plt
5 from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
6 import threading
7 import csv
8 import datetime
9 import time
10
11 # Function to establish serial connection with Arduino
12 def connect_to_arduino():
13     while True:
14         try:
15             ser = serial.Serial('COM4', 9600) # Adjust COM port accordingly
16             return ser
17         except serial.SerialException:
18             print("Serial connection failed. Retrying in 2 seconds...")
19             time.sleep(2)
20
21 # Function to read data from serial port
22 def read_data():
23     while True:
24         try:
25             data = ser.readline().decode('utf-8').rstrip()
26             if data and data[0].isdigit(): # Check if the line starts with a digit
27                 humidity, temperature = map(float, data.split())
28                 update_gui(humidity, temperature)
29                 update_plot(humidity, temperature)
30                 log_data(humidity, temperature)
31         except Exception as e:
32             print("Error:", e)
33             break
34
35 # Function to update GUI with temperature and humidity readings
36 def update_gui(humidity, temperature):
37     humidity_label.config(text=f'Humidity: {humidity}%')
38     temperature_label.config(text=f'Temperature: {temperature}°C')
39
40 # Function to update plot
41 def update_plot(humidity, temperature):
42     temperature_data.append(temperature)
43     humidity_data.append(humidity)

```

Fig. 4: Snippet of the Python code

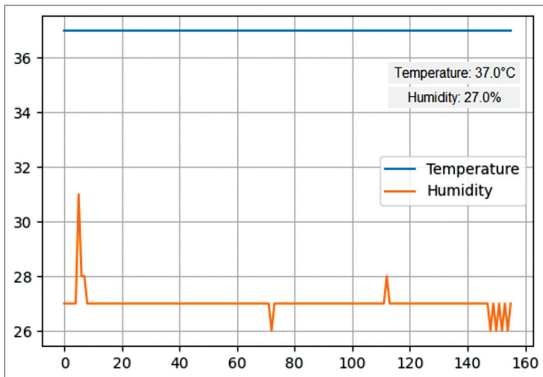


Fig. 5: GUI for real-time temperature and humidity visualisation

updates a real-time plot of temperature and humidity levels in a graphical user interface (GUI). Additionally, the Python script logs the received data along with timestamps into a CSV (comma-separated values) file, providing a structured record of environmental conditions over time.

The plotter provides users with

valuable insights into temperature and humidity trends, facilitating informed decisions and proactive environmental management. Whether used for home climate monitoring, agricultural applications, or educational purposes, the real-time environmental data logging system presented here offers versatility and practicality.

Two codes are used in this system. The first code runs natively on the Arduino, and the other code runs on

EFY Note
The source code
of this project
is available

for free download at
www.electronicsforu.com

	A	B	C
1	06-04-2024 12:16	24	36
2	06-04-2024 12:16	24	36
3	06-04-2024 12:16	24	36
4	06-04-2024 12:16	24	36
5	06-04-2024 12:16	24	36
6	06-04-2024 12:16	24	36
7	06-04-2024 12:16	27	36
8	06-04-2024 12:17	30	36
9	06-04-2024 12:17	25	36
10	06-04-2024 12:17	25	36
11	06-04-2024 12:17	24	36
12	06-04-2024 12:17	24	36
13	06-04-2024 12:17	24	36
14	06-04-2024 12:17	24	36
15	06-04-2024 12:17	24	36
16	06-04-2024 12:17	24	36
17	06-04-2024 12:17	24	36
18	06-04-2024 12:17	24	36
19	06-04-2024 12:17	24	36
20	06-04-2024 12:18	24	36
21	06-04-2024 12:18	24	36
22	06-04-2024 12:18	24	36

Fig. 6: Data logging in CSV file

a laptop or single-board computer like the Raspberry Pi. Upload the Arduino code first, then in the Python code on the SBC like the Raspberry Pi or laptop, check the COM port of the Arduino and configure the Python code. The Tkinter library for Python needs to be installed if it is not installed, using the terminal. Fig. 3 shows the snippet of the source code written on the Arduino platform while Fig. 4 shows the snippet of the Python code.

Construction and testing

After uploading the source code onto the Arduino Nano board, interconnect the DHT sensor as shown in Fig. 2. Connect the device to a laptop or SBC and then run the Python code. It will show the sensor data graph and save the data in a sheet. The sensor data can be opened and viewed in the sheet. Fig. 5 shows the GUI for real-time temperature and humidity visualisation while Fig. 6 shows the data logging in a CSV file. **EFY**



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